

## Addendum: Exact Evaluation of Dual-Marker Specificity

After writing the original note, I revisited the optimization setup and found that the dual-marker specificity objective can be evaluated exactly, without relying on a differentiable relaxation.

Let  $X \in \{0, 1\}^{N \times G}$  be the binarized single-cell expression matrix with  $N$  cells and  $G$  candidate surface genes. Cells are partitioned into lesion and healthy subsets,  $X_L$  and  $X_H$ .

For a gene pair  $(i, j)$ , define lesion and healthy co-expression frequencies as

$$P_L(i, j) = \frac{1}{|L|} \sum_{k \in L} X_{ki} X_{kj}, \quad P_H(i, j) = \frac{1}{|H|} \sum_{k \in H} X_{ki} X_{kj}.$$

The specificity score is then

$$S(i, j) = P_L(i, j) - P_H(i, j).$$

The key point is that all pairwise co-expression frequencies are obtained at once via Gram matrices:

$$P_L = \frac{1}{|L|} X_L^T X_L, \quad P_H = \frac{1}{|H|} X_H^T X_H.$$

Entry  $(i, j)$  of each matrix is exactly the fraction of cells co-expressing genes  $i$  and  $j$  in that population, so the full specificity matrix is simply

$$S = P_L - P_H.$$

The globally optimal pair follows directly from

$$(i^*, j^*) = \arg \max_{i \neq j} S(i, j).$$

With  $G \approx 3000$ , this evaluates roughly 4.5 million unordered pairs using two matrix multiplications, making exact search computationally straightforward.

This does not contradict the differentiable formulation in the main note. Empirically, that optimization still converges to near-optimal solutions across random seeds, often recovering the second-best exact pair. This suggests a landscape with shallow local minima that still guides the search toward highly enriched targets.

### Example Candidate Pairs

Exact enumeration on the current dataset yields the following top-scoring combinations:

| Marker pair    | Specificity score |
|----------------|-------------------|
| GAS1 + LRP1    | 0.579             |
| BST2 + GAS1    | 0.534             |
| ATRAID + GAS1  | 0.514             |
| CD164 + GAS1   | 0.500             |
| DDR2 + GAS1    | 0.481             |
| BST2 + LRP1    | 0.477             |
| IL6ST + SGCE   | 0.470             |
| PRNP + SGCE    | 0.470             |
| BST2 + IL6ST   | 0.470             |
| GAS1 + SERINC1 | 0.468             |

Several high-scoring pairs repeatedly include *GAS1*, *BST2*, *LRP1*, and *SGCE*, supporting the view that these genes mark stable lesion-associated surface programs in this dataset.

Overall, the exact formulation offers a simple and efficient full-pair evaluation, while the differentiable formulation remains useful as a geometric perspective and a path to richer objectives (for example, higher-order marker sets or additional safety constraints).